

ECEN 607: Advanced Analog Circuit Tech

**Lab6: a) Multistage Op-AMP with NMC
b) Using Cascoded Miller Compensation**

Lab-Report

Name: Harshit Gupta

UIN: 735006036

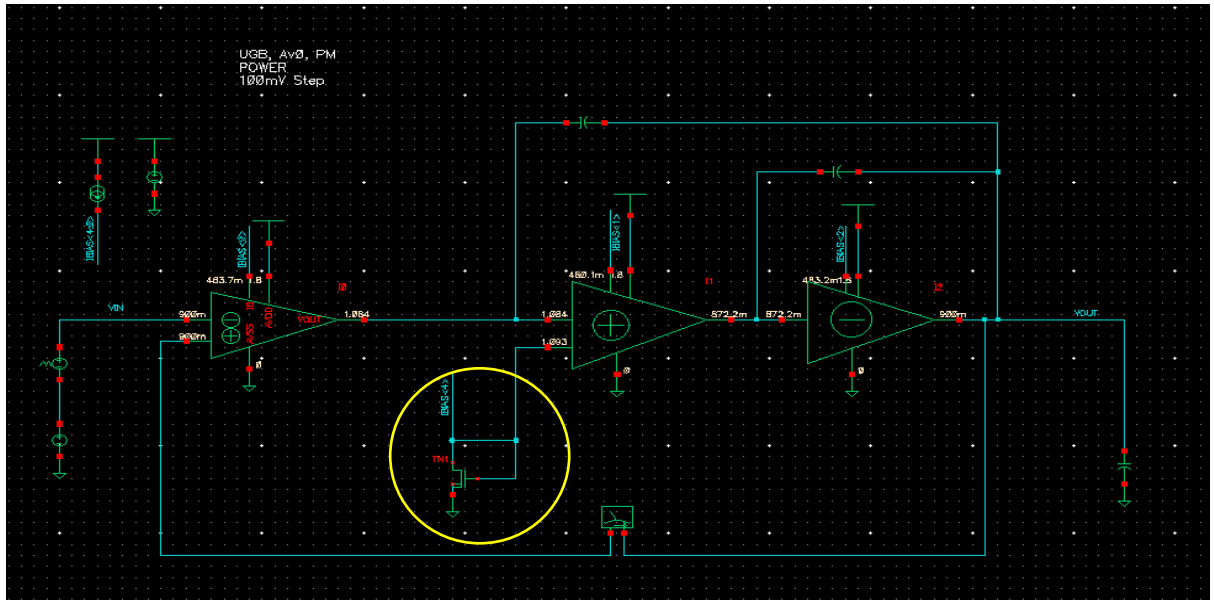
Section: 602

NESTED MILLER OPAMP

1. Prelab Design Simulation:

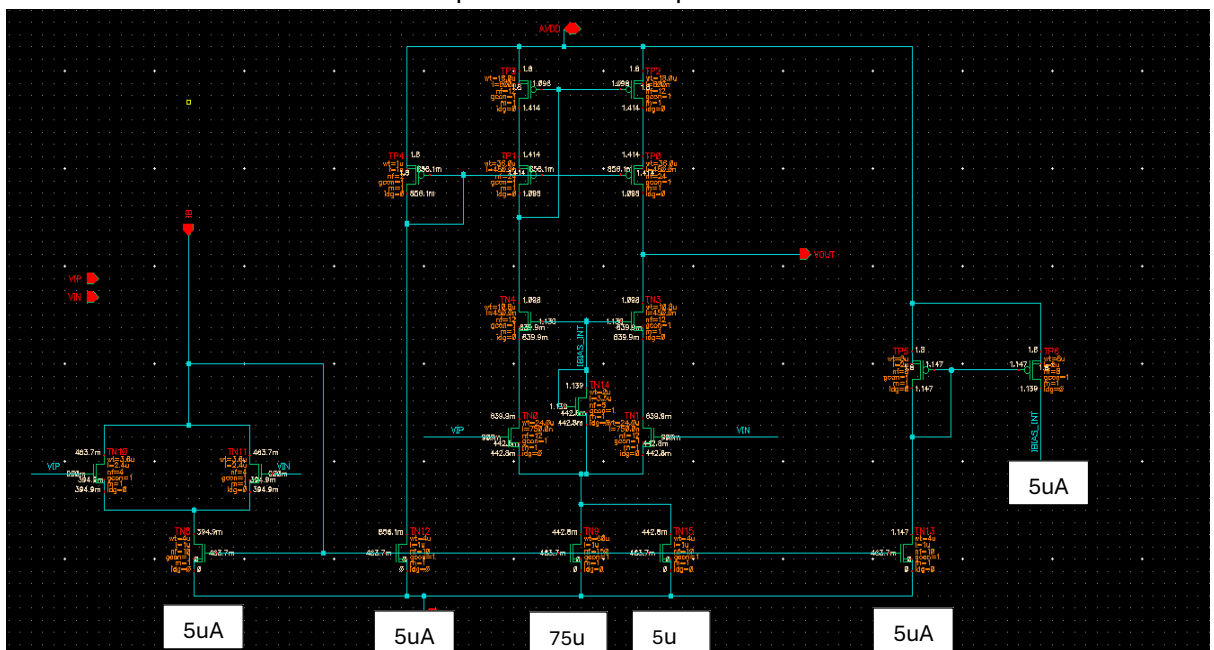
Overall Schematic (in Closed-Loop): Only 5uA ideal current source is used as was asked in the manual

In the second stage there was a requirement for common-mode biasing voltage which was provided by diode connected transistor as encircled below.

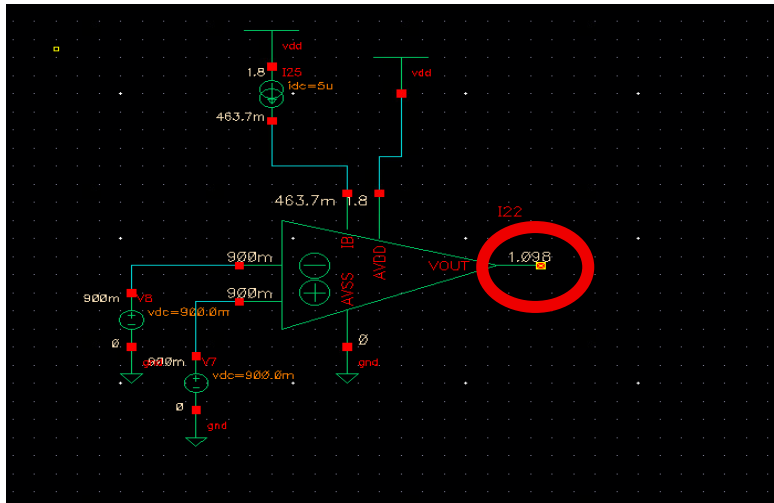


There are 3 stages in the NMC opamp, all the three stages are verified below (appropriate operating points, gain) were checked.

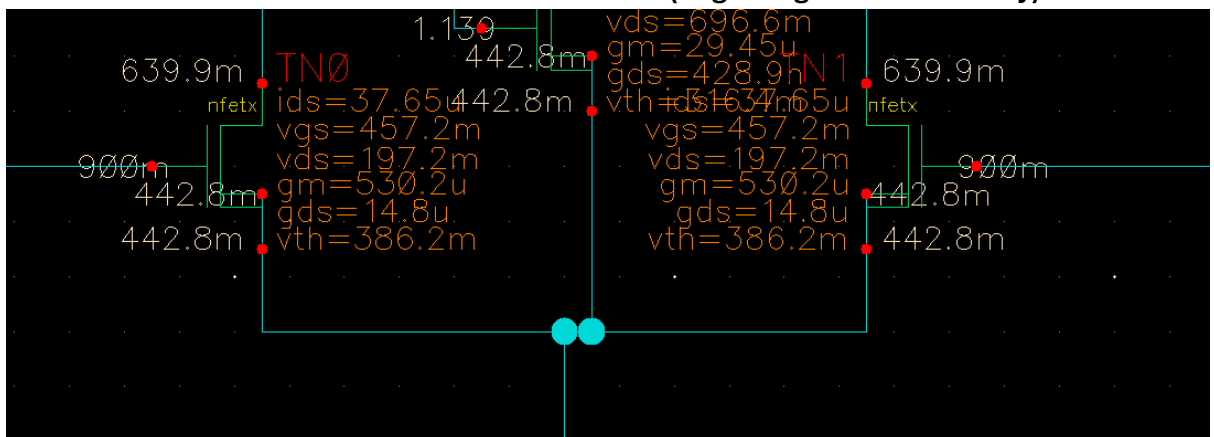
Stage-1 (Telescopic Cascode): Differential Input & Single-Ended Output (All the currents are properly marked below). Biasing circuits are also shown. Only 5uA ideal current source is assumed which is provided at the top-level.



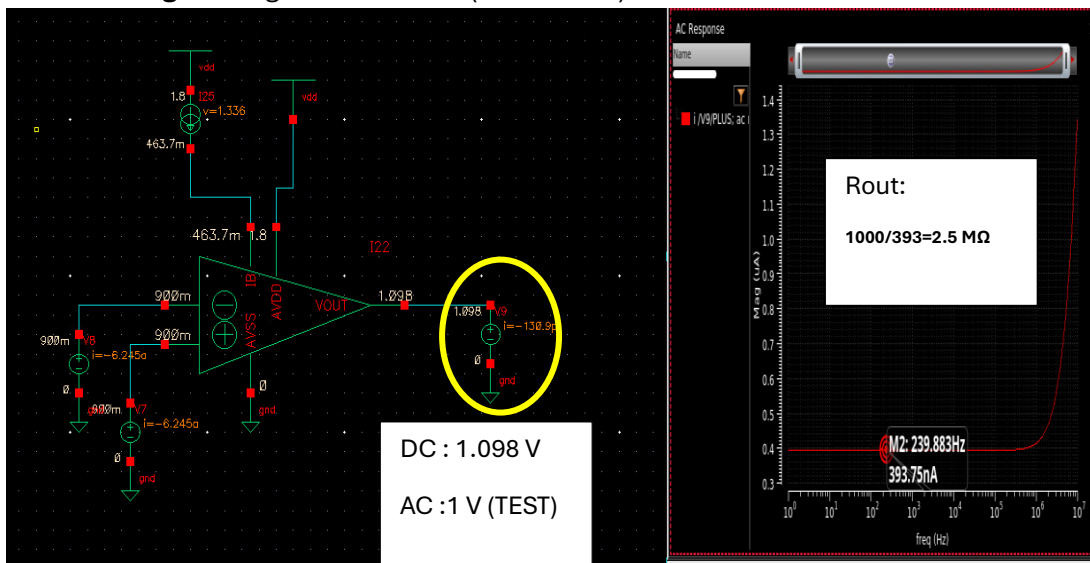
Important Node Voltages: V_{out} in balanced case is 1.098 Volts, this is required for the second stage design. (1.098 Volts is as expected in the prelab)



Operating Pts → Input G_m : 530uS (which is higher than g_m mentioned in prelab) which means there will be a boost in U_{GB} number (might degrade the stability).



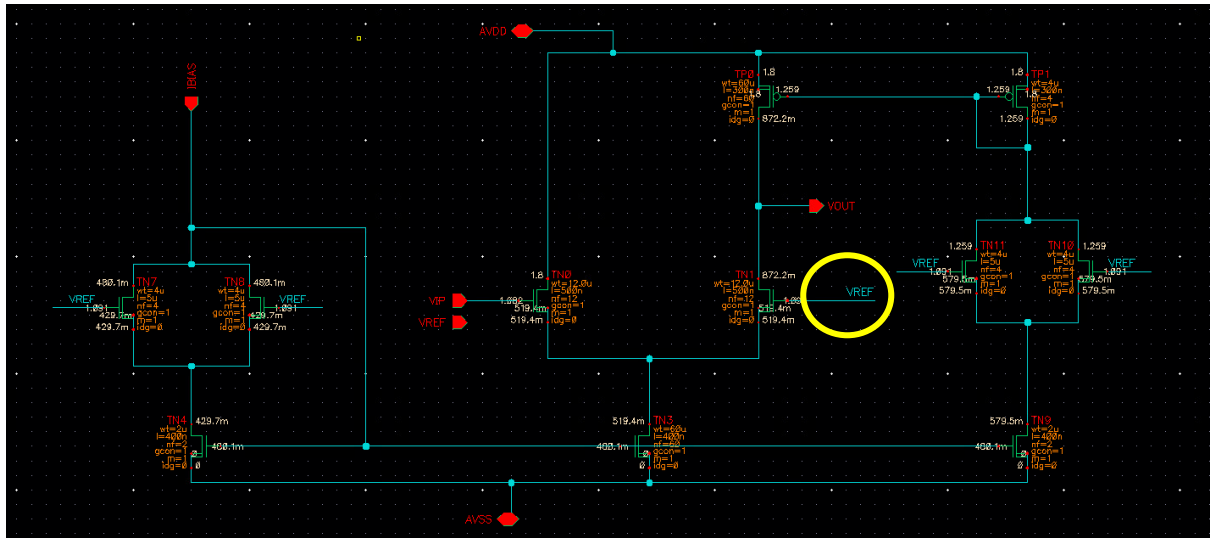
Rout-1stStage: Using AC simulation (2.5 Mohms)



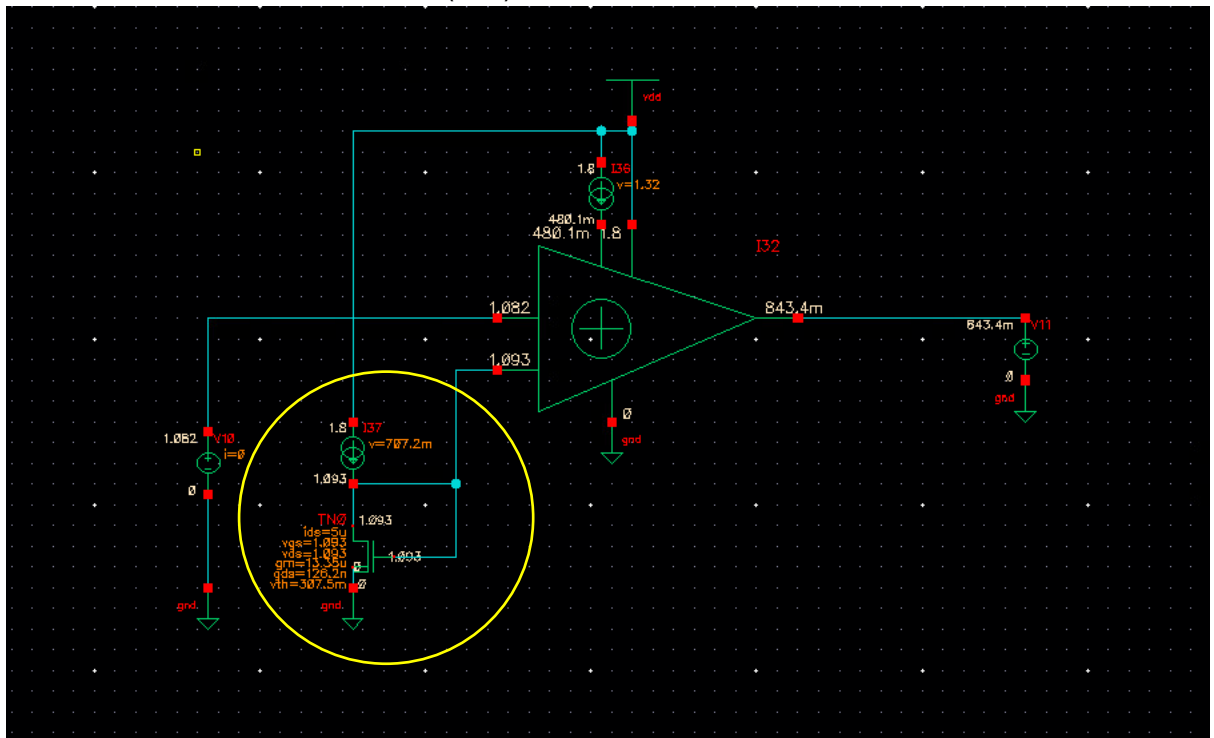
$A_{v_STAGE1} = G_m \cdot R_{out} = 1346 = \mathbf{62.5\text{ dB}}$ (less than what was mentioned in the prelab) but should be good since the design had sufficient margins.

Stage-2 (non-inverting amplifier): Single-Ended i/p & o/p

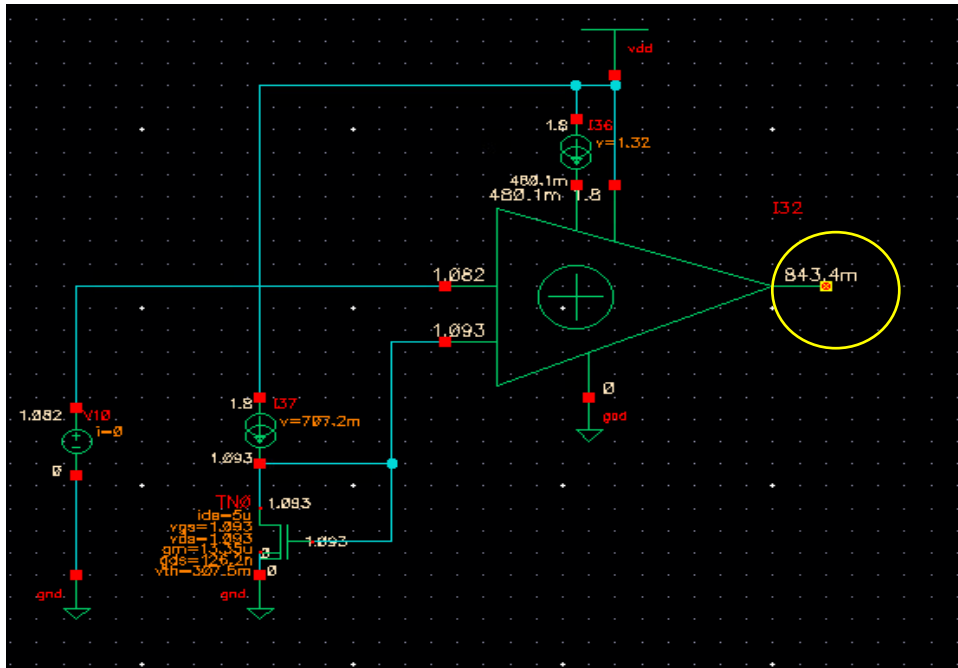
Source-Follower + Common Gate in cascade (like folded-Cascode but single ended)



Please Note: Signal is fed at the VIP (biased by the output common mode of the first stage). The bias of common-gate is just a DC voltage source (no-signal), as encircled above. As shown below even the reference for this is generated using diode-connected transistor and ideal current source (5uA)



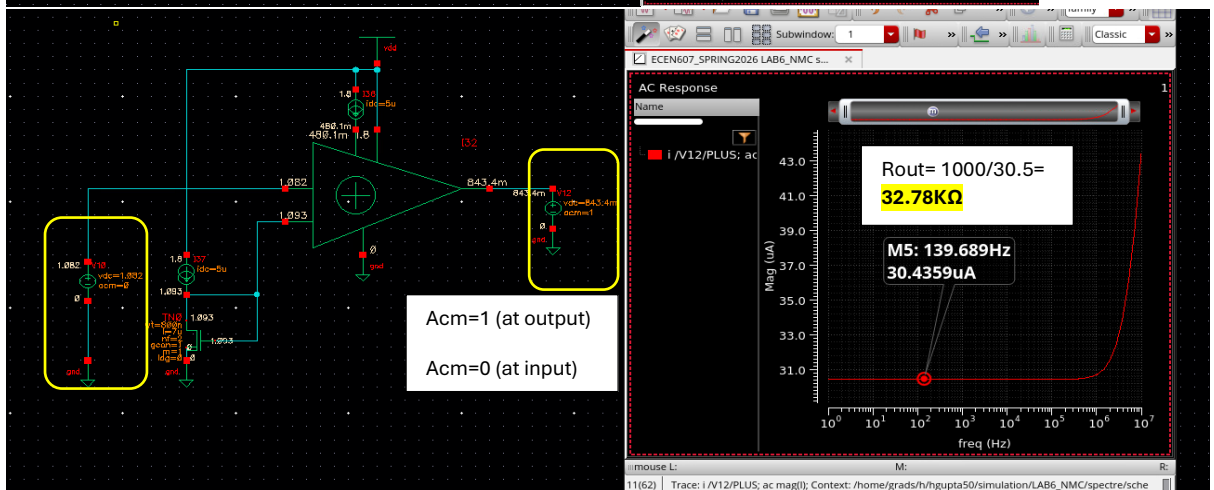
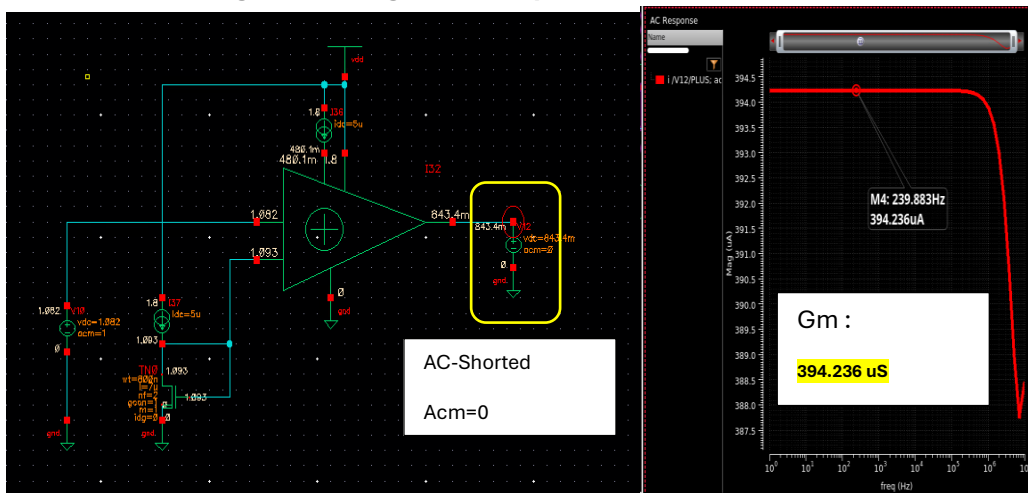
$V_{out}=843.4\text{mV}$ (this is required for the design for the 3rd stage common-source amplifier).



Gm & Rout: Using AC sim

For Gm → Output was AC shorted and input small signal was applied

For Rout → No signal at the input (AC-shorted) and Test AC Voltage was applied without disturbing the biasing at the output.

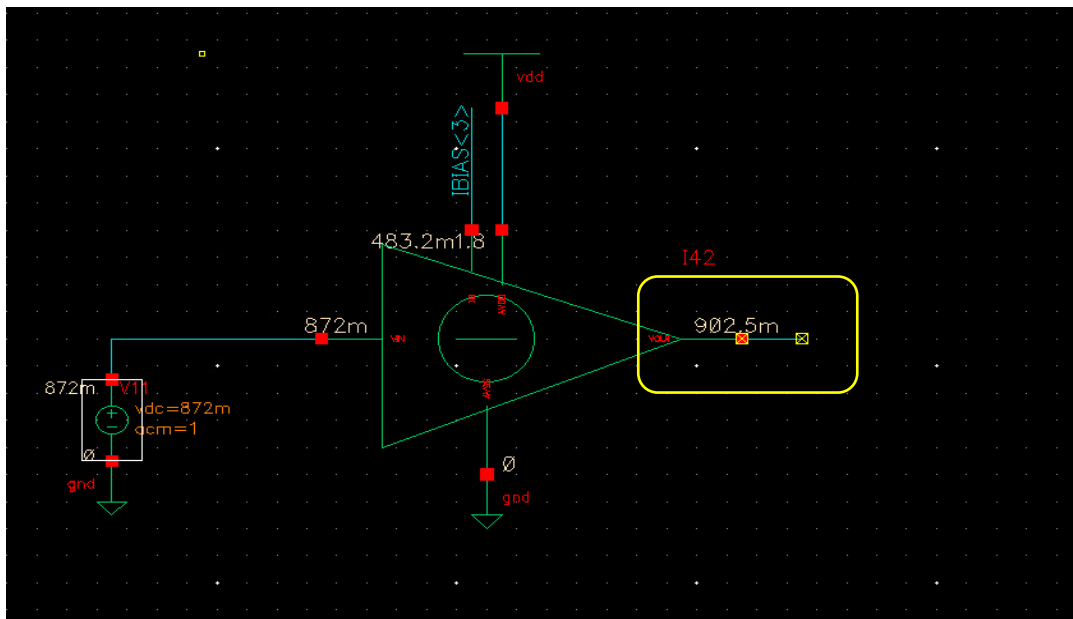
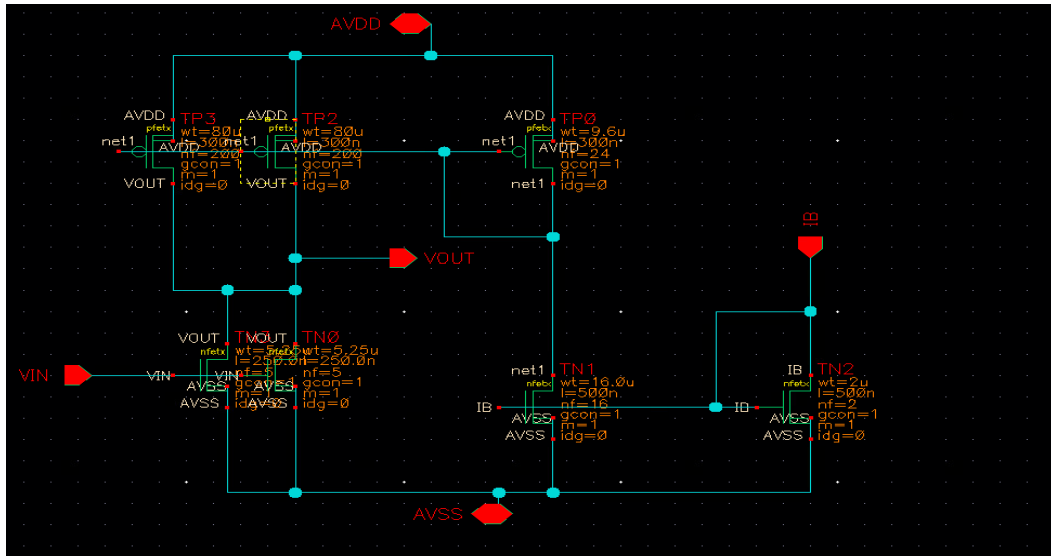


Gm2 = 394.236uS (very close to prelab value)

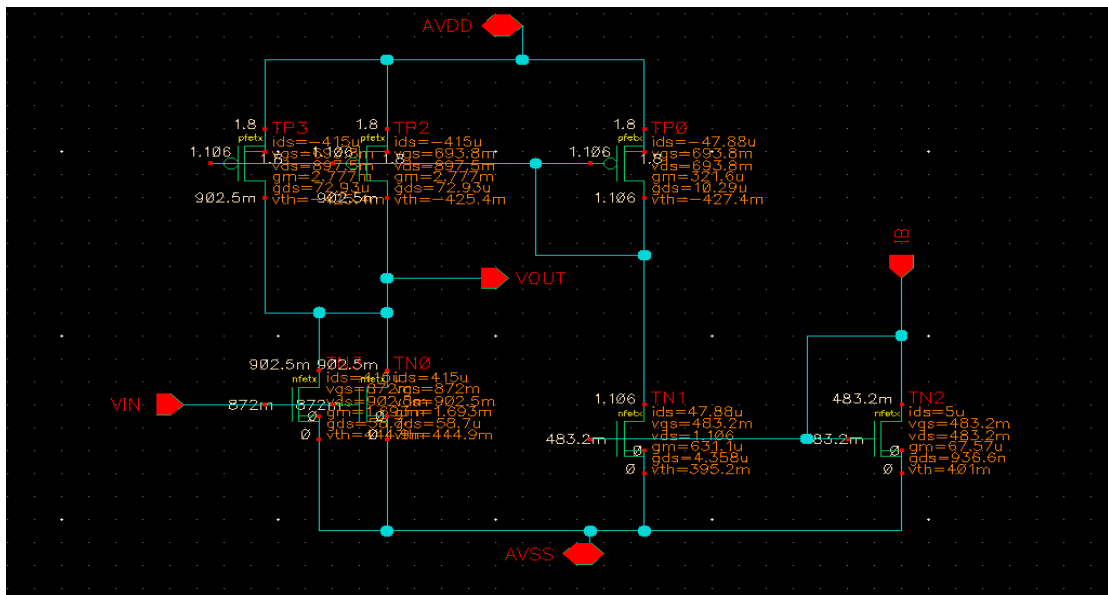
Rout2 = 32.78 Kohms

Av2 = 12.6 = 22 dB

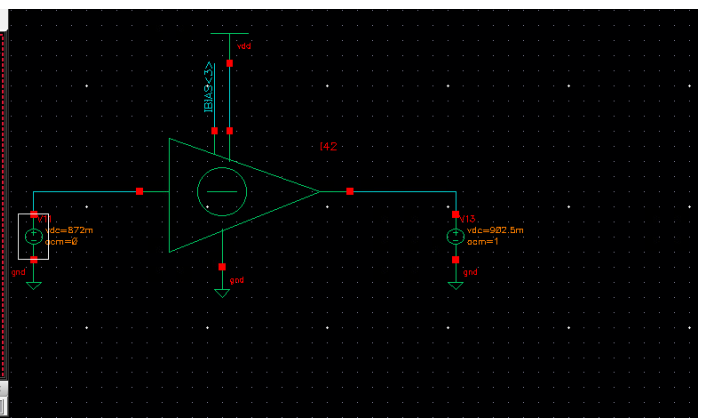
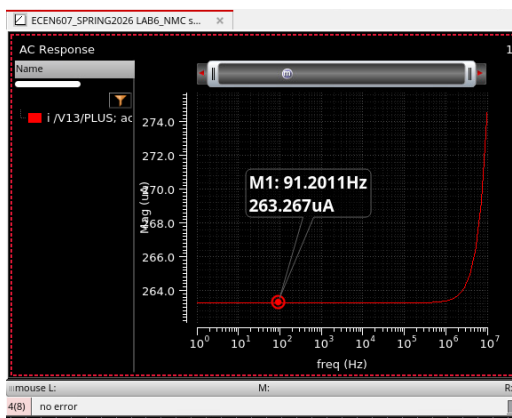
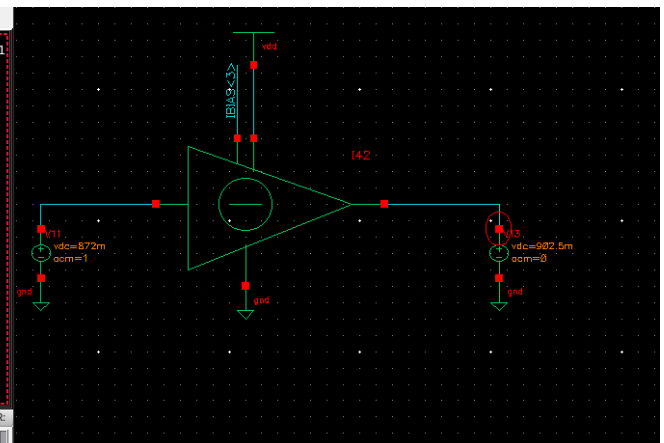
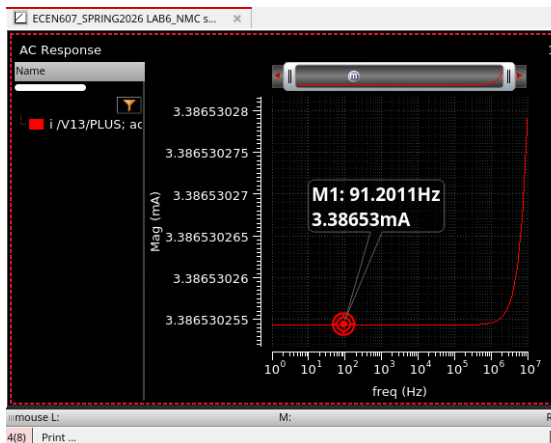
Stage-3 (Simple Common-Source Stage):



Currents were properly matched in the output stage, so the output was properly sitting at mid-rail.



Gm & Rout Calculations:

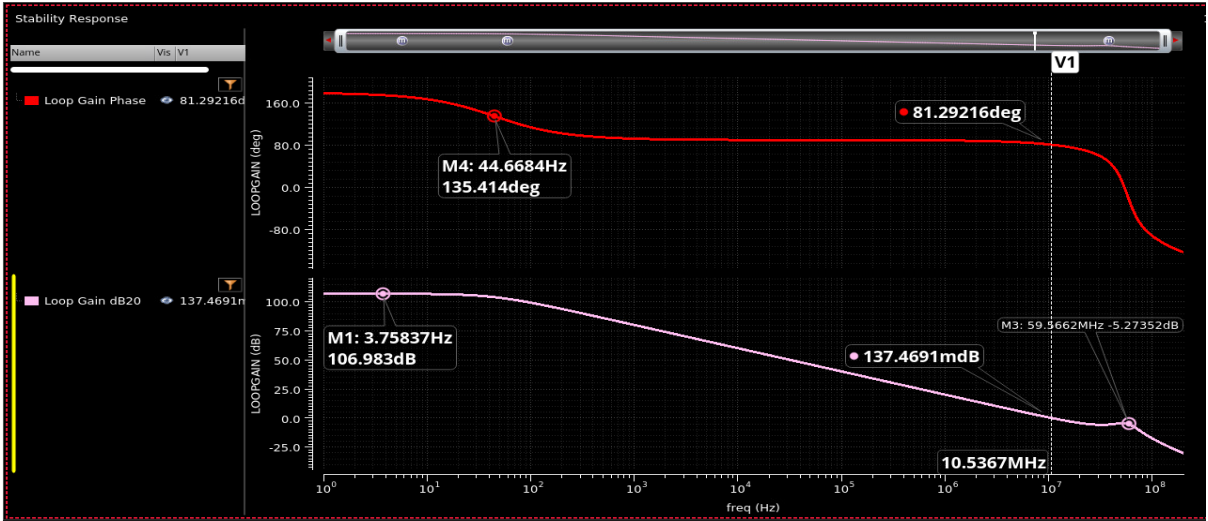


$Gm3 = 3.38 \text{ mS}$
 $Rout3 = 3.8 \text{ K}\Omega$
 $Av3 = 12.84 = 22.17 \text{ dB}$

$Av_{Total} = 62.5 + 22 + 22.17 = 106.67 \text{ dB}$ (will be confirmed below on the top-level simulation)

Please Note: It was ensured in the design that inner-most loop is the fastest, due to which most power is consumed in the 3rd Stage (common-source stage), as to get higher g_m more current was burnt (more than 90% power was consumed in the 3rd stage).

a. A_{v0} , PM, UGB, Power (closed-loop configuration):



A_{v0} : 106.98 dB (>100dB) → Marker M1

PM: 81.29 degrees (> 50 degrees) → Vertical Marker V1

UGB: 10.53 MHz (>5MHz) → Vertical Marker V1

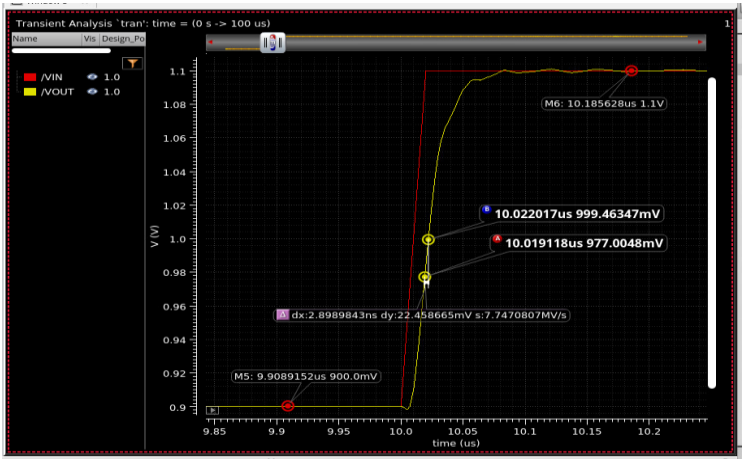
Complex-poles arriving at 60MHz, it was ensured that peaking was controlled (less than 0 dB horizontal line)

dc_stb_tran	lgnd	expr	(IS("TN1/S") + IS("I0/AVSS") + IS("I1/AVSS") + IS("I2/AVSS"))	point	☒	☐	
dc_stb_tran	Power_consumed	expr	abs((1.8 * lgnd))	point	☒	☐	< 2m
dc_stb_tran	lgnd		-1.168m				
dc_stb_tran	Power_consumed		2.103m				< 2m
							near

Power Consumed: 2.103mW (slightly above 2mW spec)

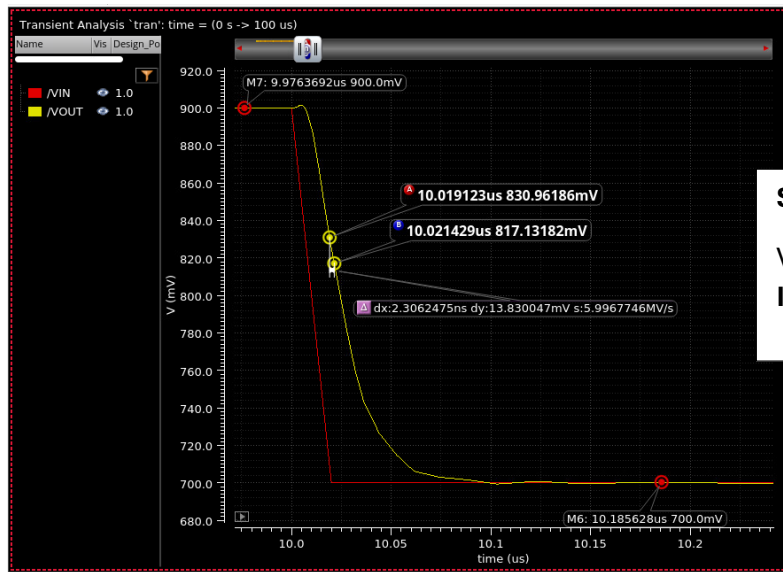
b. Slew-Rate (Closed-loop, fast step at the input):

Ramp_rate: 10MV/s, input step of 200mV was given. Since the slew rate is less than 10MV/s, output won't be able to follow input at this value and output rate of change will be limited by the slew rate.



SR+ = 7.74MV/s

Value close to I_{bias_stage1}/C_{load} .



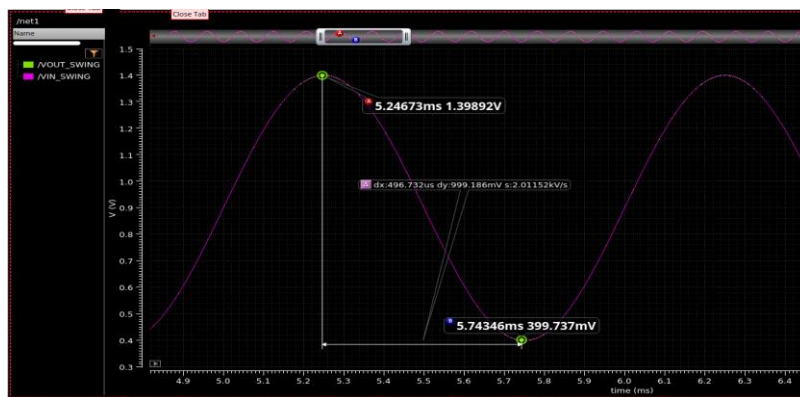
SR+ = 5.99 MV/s

Value close to
Ibias_stage1/Cload.

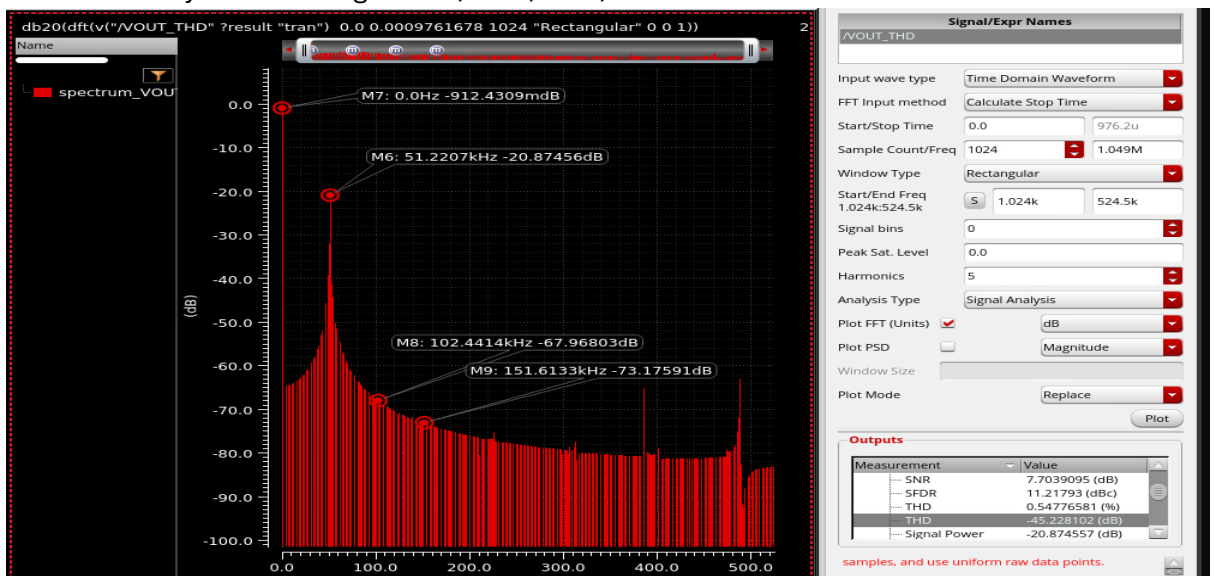
c. Output Swing (Closed-loop, with low-frequency sinusoidal input):

$$V_{IN} = 900\text{mV} + 0.5\sin(2\pi \cdot (1000) \cdot t)$$

Dominant pole is at 47 Hz, so the signal will be attenuated a bit, but still gain is higher than 60dB even at 1kHz.



2. OPAMP in unity-buffer configuration, HD2, HD3, THD:

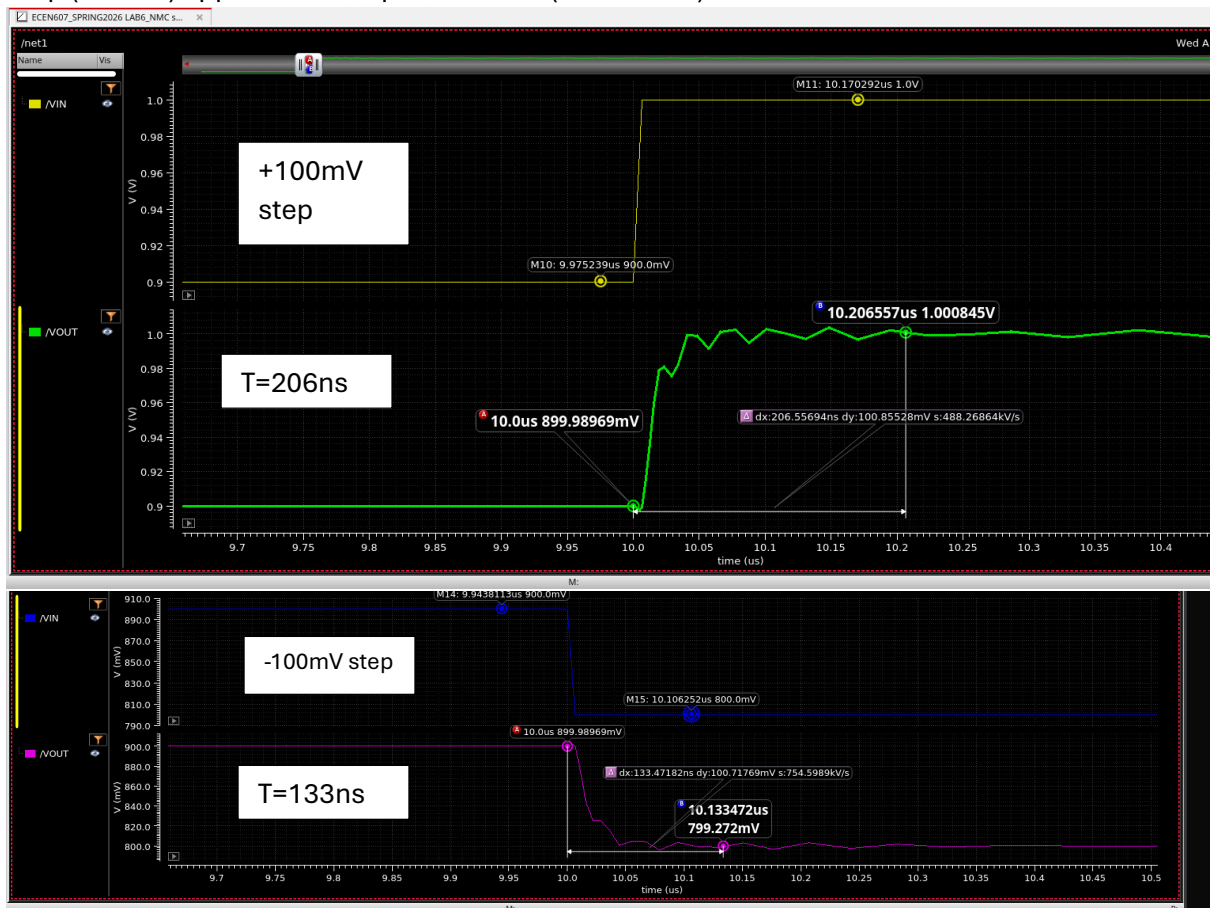


HD2 = -47dBc
 HD3 = -53 dBc
 THD = -45.22 dB

3. Step-response of Opamp:

0.1% accuracy (7*tau) need to be figured out.

Step (100mV) applied with slope of 15MV/s (> slew-rate)



Some small rings can be seen in the settling behaviour even after having a phase margin of 80 degrees. This can be associated with complex-conjugate poles which arises due to the inner loop.

7*tau settling ~ (7/4.6) * (1/f_ugb) ~ 1.52 * (1/10M) ~ 152 ns (close to what is seen above).

4. Comparison Table:

Parameters	SPEC	Achieved Sim Result
Av0	100 dB	106 dB
PM	>50	81.29 degrees
Swing	1 Volts	1 Volts
UGB	5MHz	10.5MHz
Power	2mW	2.1mW
Slew Rate	>5MV/s	6MV/s

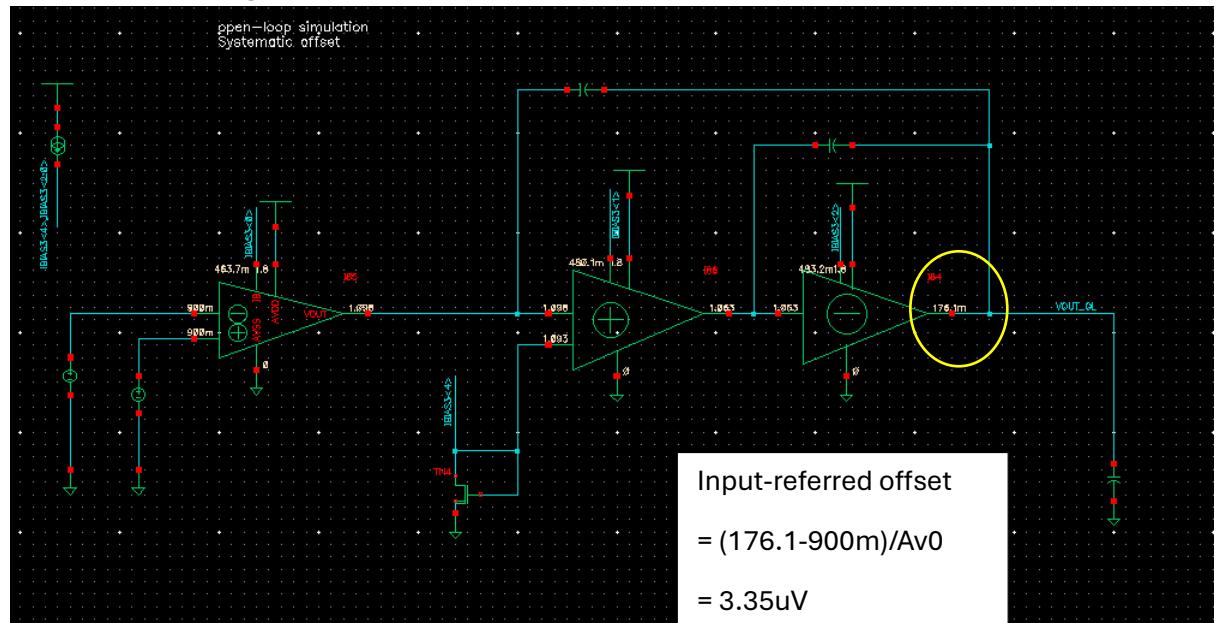
5. Final Comments, Conclusion:

NMC requires very precise control of the complex-conjugate poles. But once they are shifted away, it is easy to achieve high phase margins.

The NMC compensation is possible only if the inner-most loop is fast which requires power to be burnt in the 3rd stage. (some complex schemes which can boost the gm will be very helpful over there).

Input-offset Voltage:

OPEN-LOOP Configuration:



Ahuja Compensation (Cascode-Miller Compensation)

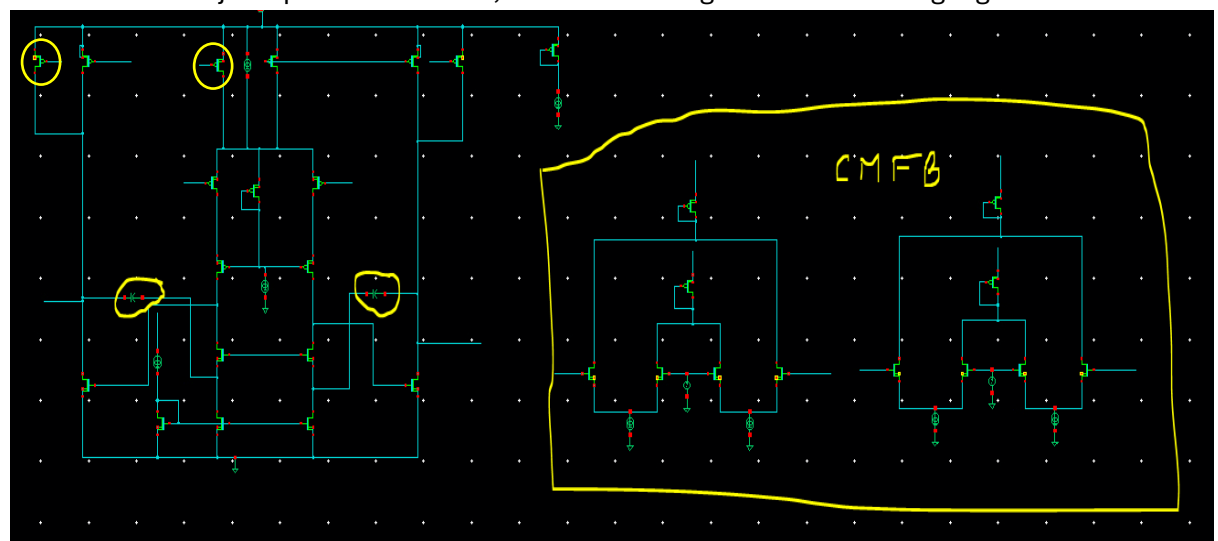
6. Amplifier Design and its simulations:

First Stage: Telescopic Cascode

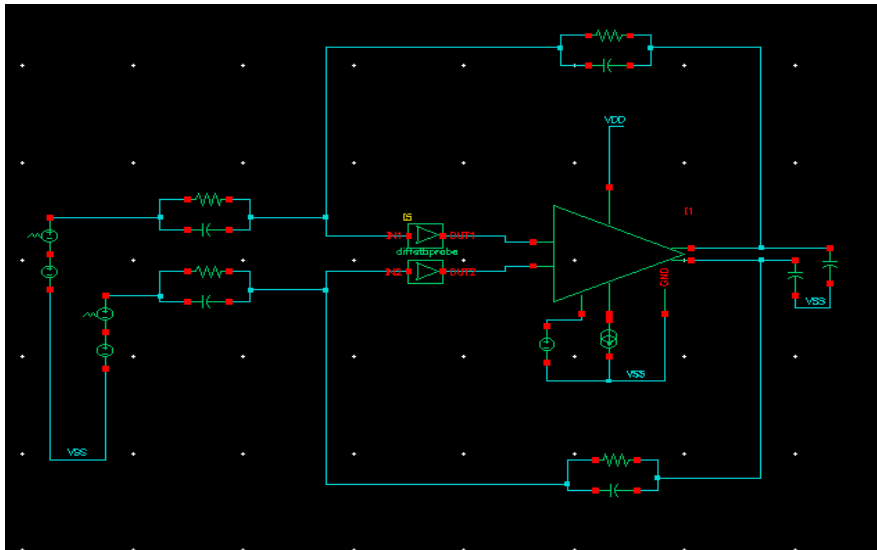
Second Stage: Common-Source Stage

Both stages are differential which implies both the stages requires common mode feedback structure.

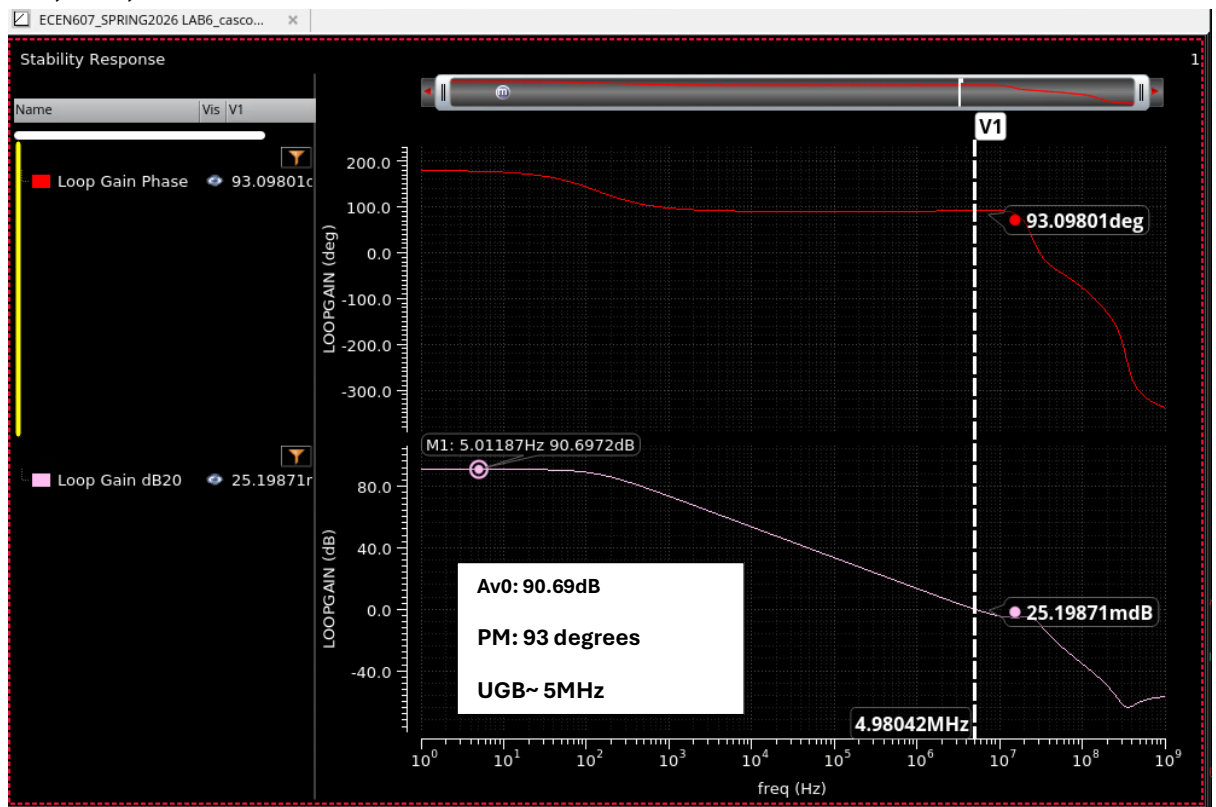
Schematic: Ahuja Cap & CMFB circuit, cmfb controlling transistors are highlighted.



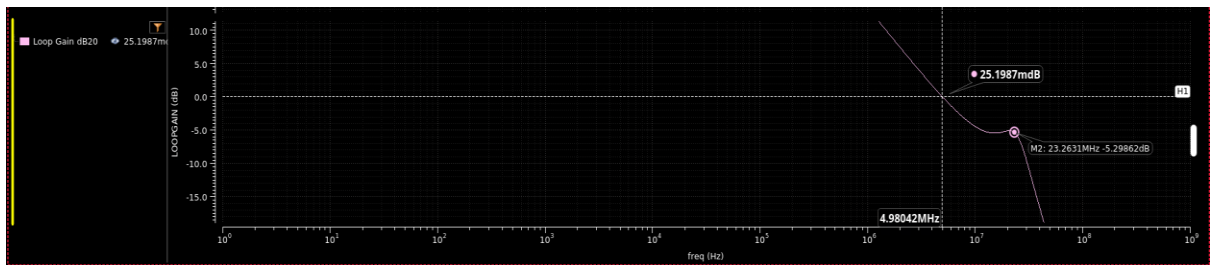
TESTBENCH Used:



Av0, UGB, PM:

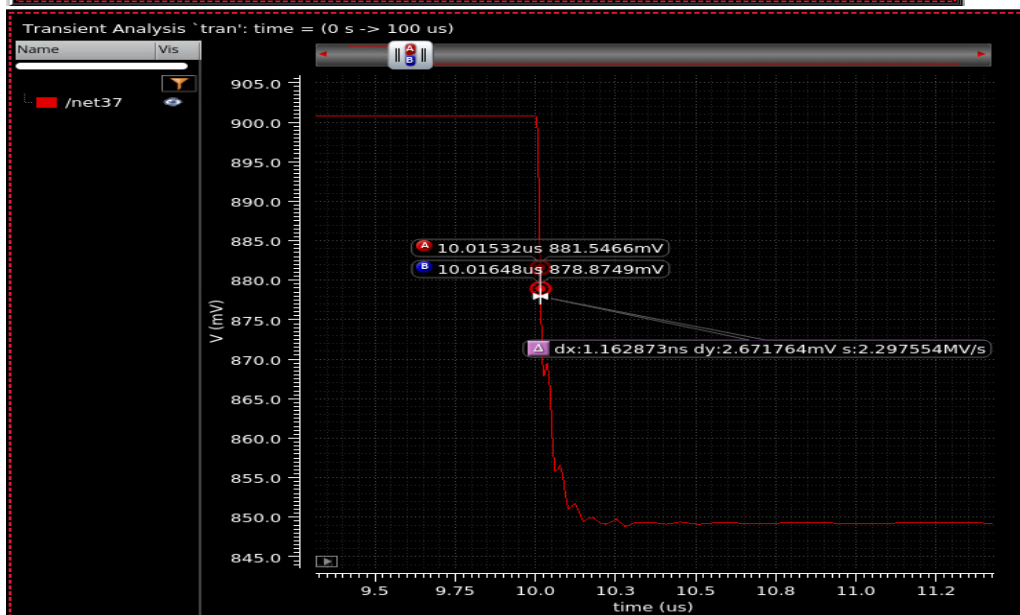


Peaking was seen at high frequency due to complex conjugate poles, but it was ensured that peaking doesn't let the magnitude plot cross the 0dB (end up in multiple ugb). As can be seen below. The was possible by having large value of "gm" for the cascode transistor.

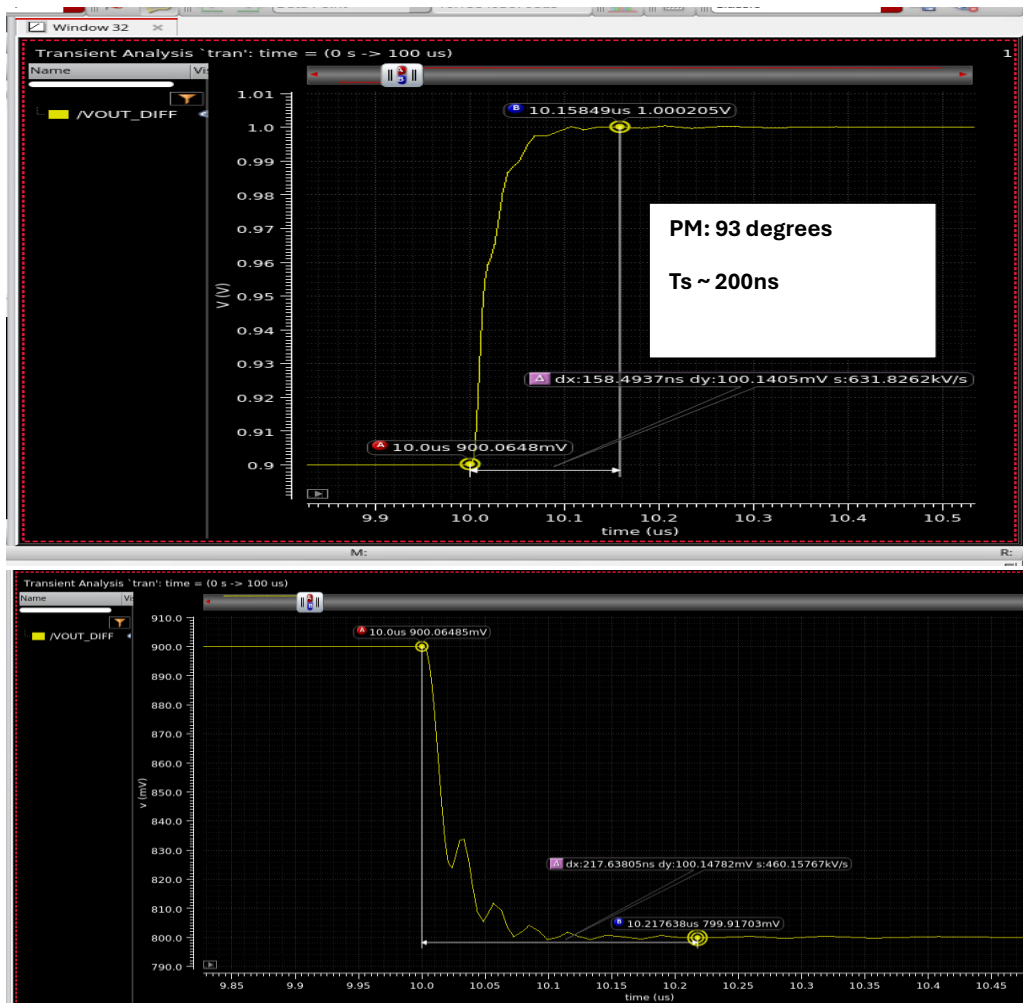


SlewRate:

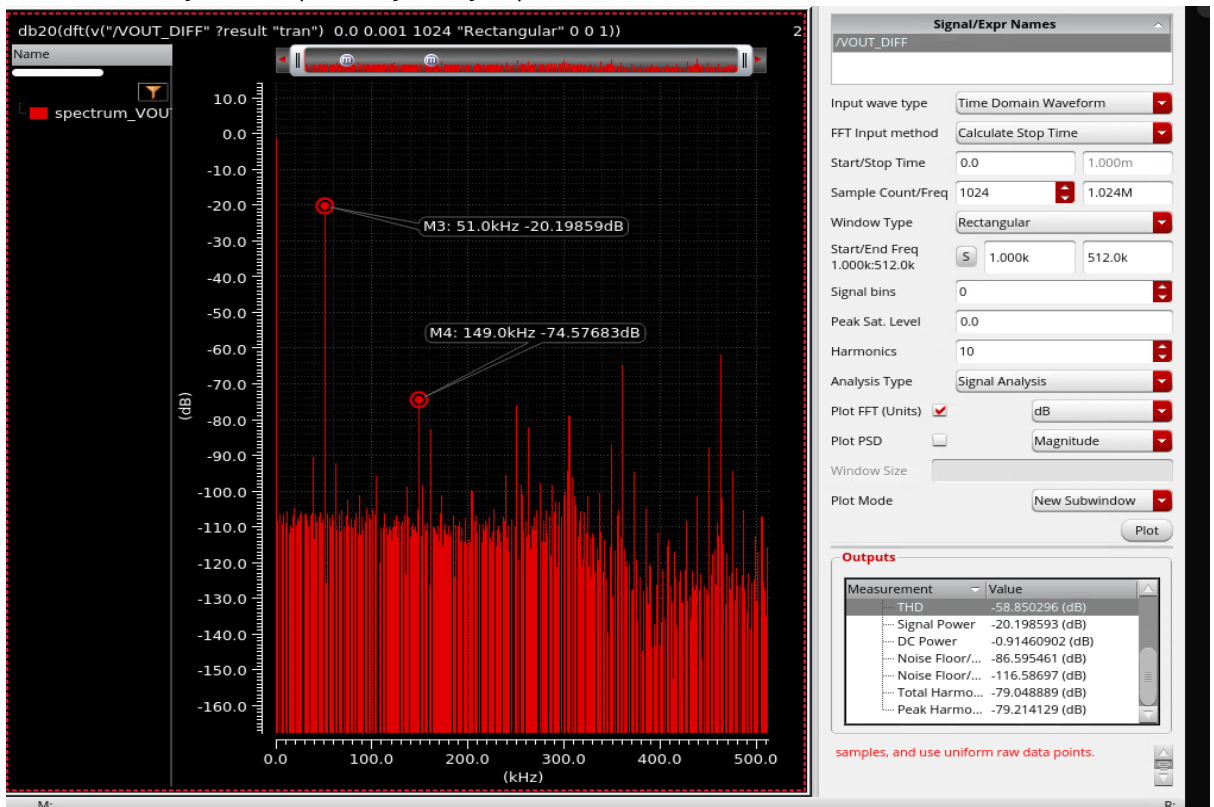
Fast-Step was given at the input



Step Response (+/- 100mV step): 100mV differential steps (+50mV & -50mV)



OPAMP in unity -buffer (linearity Analysis):



Comparison Table & comments → **provided in part-10**

7. Comparison with Miller:

Key difference between Miller and cascode-miller

- 1) PSRR (+): In miller at the high frequencies miller cap act like a short-circuit and supply ripples are transferred to output unfiltered, which degrades the supply rejection at high frequency. This is not the case in the cascoded-miller as there is isolation provided with the help of cascode-transistors.
- 2) Amount of miller cap required: Amount of miller cap required in cascoded miller is lesser as the non-dominant pole is at higher location and there is no RHP zero (atleast at frequencies comparable to non-dominant pole).

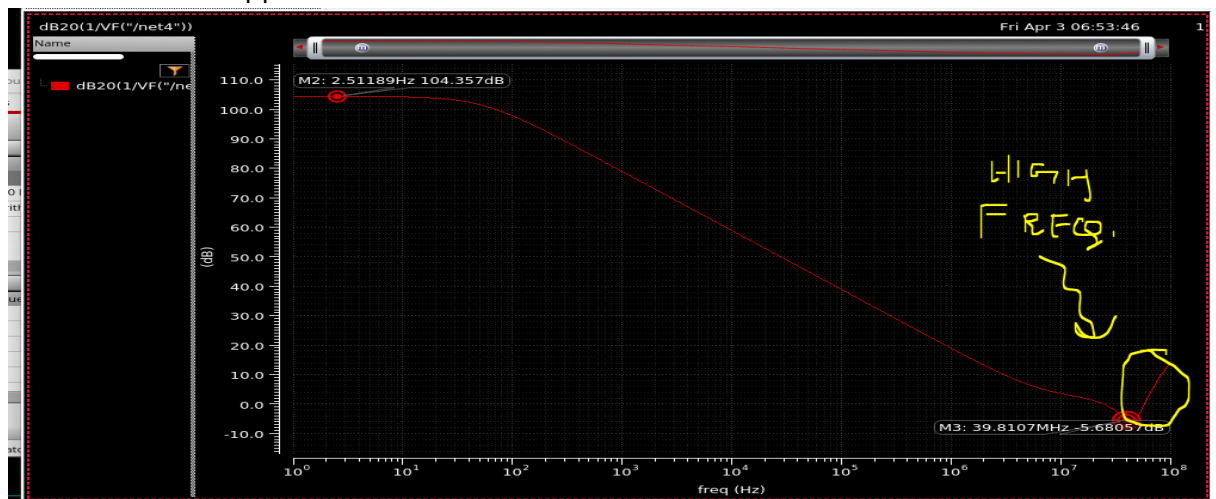
8. 100mV pulse response:

Done in part-6

9. (+/-) PSRR plots, input offset voltage, input referred noise (10Hz - GBW)

PSRR plot (+): VDD rejection

At VDD ac=1 was applied with VDC=1.8 Volts



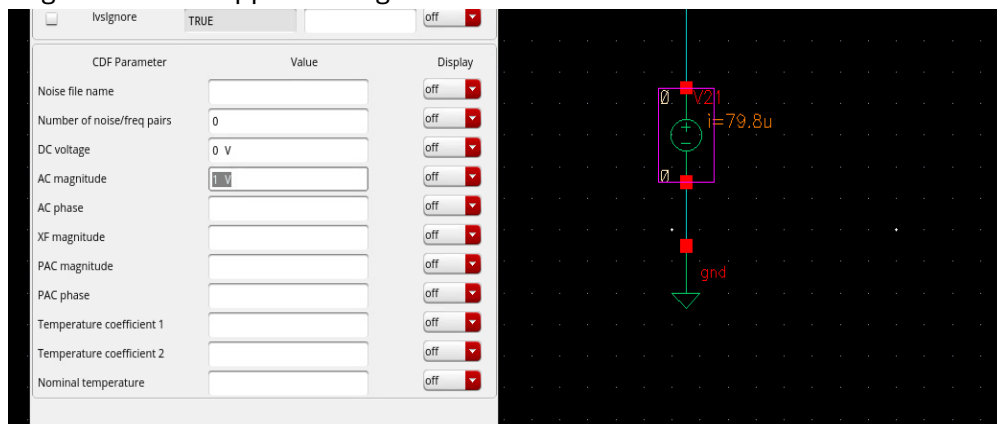
There are 3 sections in the above plot.

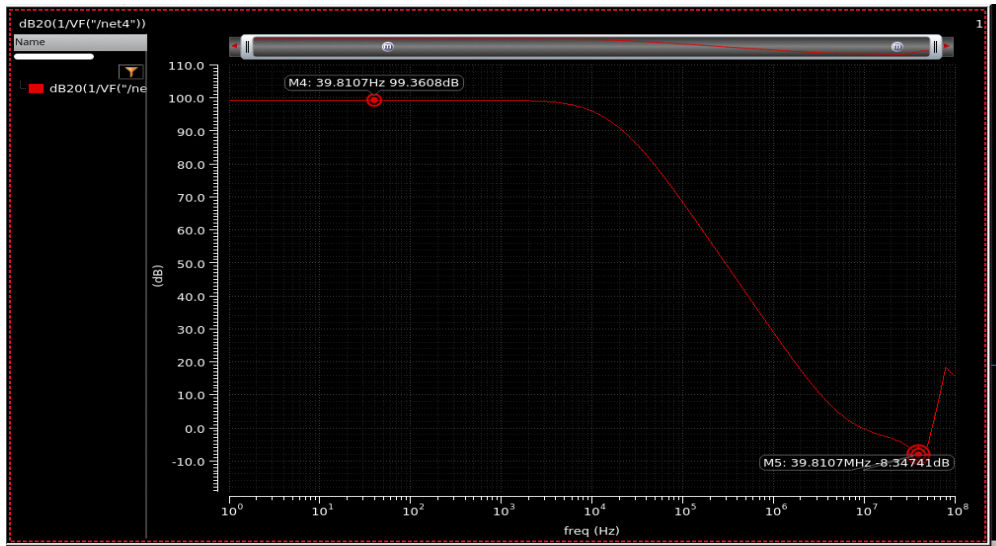
At very high frequency → output cap becomes dominant impedance (making impedance closer to lower value). Which implies lesser ripple at the output.

At lower frequency → PSRR mainly dictated by the loop gain

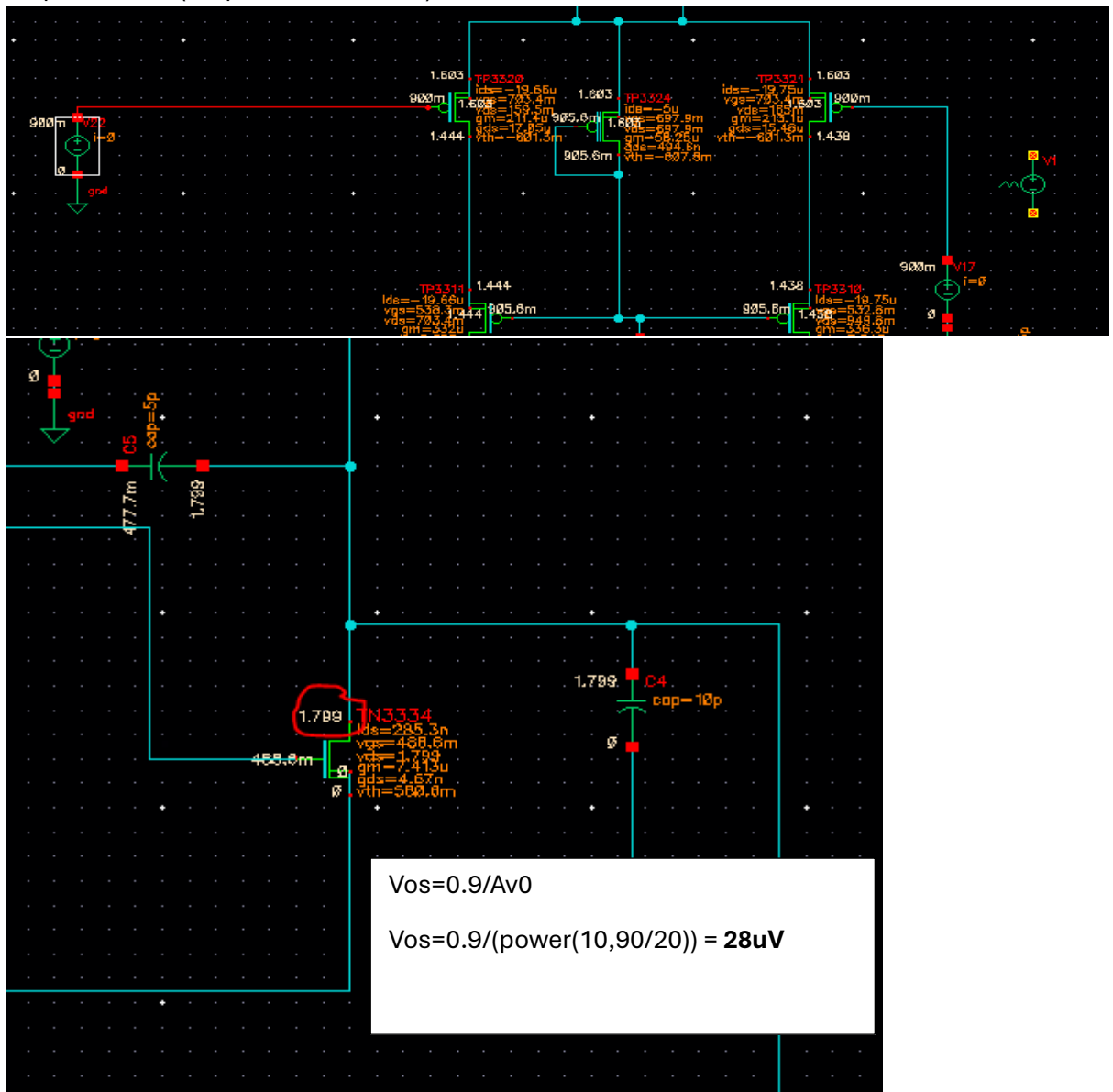
PSRR plot (-): GND rejection

At gnd ac=1 was applied alongside VDC=0





Input-offset voltage: Balanced inputs were applied and output common-mode was checked without CMFB circuit, since we want to characterize the offset due the amplifier core. (output CM – mid_rail)/Av0



Device	Param	Noise Contribution	% Of Total
/M2	Sf1	1.70849e-05	15.06
/M1	Sf1	1.70801e-05	15.05
/M11	Sth	1.70692e-05	15.03
/M2	Sth	1.51883e-05	11.90
/M1	Sth	1.51692e-05	11.87
/M8	Sth	1.20464e-05	7.49
/M7	Sth	1.08523e-05	6.08
/M6	Sth	1.07527e-05	5.96
/M15	Sth	1.04774e-05	5.66
/M11	Sf1	5.73165e-06	1.69

Integrated Noise Summary (in V) Sorted By Noise Contributors
Total Summarized Noise = 4.40285e-05
Total Input Referred Noise = 4.66281e-05
The above noise summary info is for noise data

Input-ref noise = 46uV rms

10. Comparison Table & Final Comments:
NMC vs Cascode-miller compensation:

	NMC	CascodedMiller
DC Gain(dB)	106	90.69
Phase Margin	81.29	93
GBW(MHz)	10.5	5
PSRR(dB)	62	104
Vos(uV)	3.35	28
Noise(uVrms)	134	46
Cc(pF)	8(Cm2) ,0.4(Cm1)	5
Clload	10pf	

Comments:

- 1) PSRR: NMC suffers from the PSRR drawback of the miller compensation as compared to Ahuja compensation (which isolates output node from the supply)
- 2) Systematic offset was lower in NMC primarily due to more gain, which is easier to achieve in 3 stages rather than 2 stages in cascoded miller
- 3) Noise was higher in NMC primarily due to more components in the circuits.